



Safe Streets and Roads for All

Calculating Average Annual Fatality Rate

This document provides instructions on how to obtain the necessary data and calculate the Average Annual Motor-Vehicle-Involved Roadway Fatalities for inclusion in the Safe Streets and Roads for All (SS4A) Action Plan Grant application. The SS4A Notice of Funding Opportunity (NOFO) requires that applicants use the most recent five years of data available in the Fatality Analysis Reporting System (FARS). As of April 2023, the 2021 FARS data has been published and 2017-2021 data should be used.

This document uses an example of **Washington County, Maryland** for collecting and calculating the data.

Downloading Fatality Data from the Fatality Analysis Reporting System

1. Go to the [Fatality and Injury Reporting System Tool \(FIRST\)](#) website to view the FARS data. At the top of the screen, select the tab for "People." Under "Select Fatality and/or Injury," select "Persons Killed in Fatal Crashes" and Select Time Frame as 2017-2021:
2. To select a county or city, first select "State" and type in or select the state. Type in or select the county OR city in the appropriate box. Note: Users cannot select both cities and counties due to potential overlapping boundaries. In the example below, Maryland is selected as the state and Washington is selected as the county.

The screenshot displays the FIRST website interface with the following sections:

- Crashes | Vehicles | People | Drivers | Occupants | Pedestrians | Pedalcyclists** (Navigation tabs)
- Select Fatality and/or Injury**
 - ☐ Persons Involved in Fatal Crashes
 - ☒ Persons Killed in Fatal Crashes
 - ☐ All Persons Involved in Motor Vehicle Crashes
 - ☐ Estimated Persons Involved in Injury Only Motor Vehicle Crashes
 - ☐ Estimated Persons Involved in Property-Damage-Only (PDO) Motor Vehicle Crashes
 - ☐ Estimated Persons Involved in Injury and PDO Non-Fatal Motor Vehicle Crashes
 - ☐ All Persons Involved in Motor Vehicle Crashes

** No Region, State, County or City is available for Injury, PDO, and All crashes data.*
- Select Time Frame**
 - Time Frame: **Years**
 - Timeline: 2007 to 2021. The period from 2017 to 2021 is highlighted in blue.
- Select State or Region**
 - ☒ State
 - Input field: **× Maryland**
 - Input field: **× Washington**
 - Input field: **All Cities**
 - ☐ NHTSA Region

** County and City selections are available only when a single State is selected.*

** Users cannot select both Cities and Counties due to potential overlapping boundaries.*

For more information on the Safe Streets and Roads for All program, please see the [SS4A website](#).

3. Click Submit.
4. Retrieve the TOTAL # of Persons Killed in Fatal Crashes: **78**, highlighted in the table below.

Persons Killed in Fatal Crashes

Year*	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	Total
2017	3	0	1	3	1	1	1	2	0	1	1	0	14
2018	2	0	2	2	0	0	1	3	1	1	2	2	14
2019	4	1	2	0	2	2	1	1	3	7	4	1	28
2020	0	1	0	0	0	0	0	1	2	4	2	0	10
2021	1	1	1	1	0	2	1	2	1	0	2	0	12
Total	10	3	6	6	3	5	4	9	7	13	9	3	78

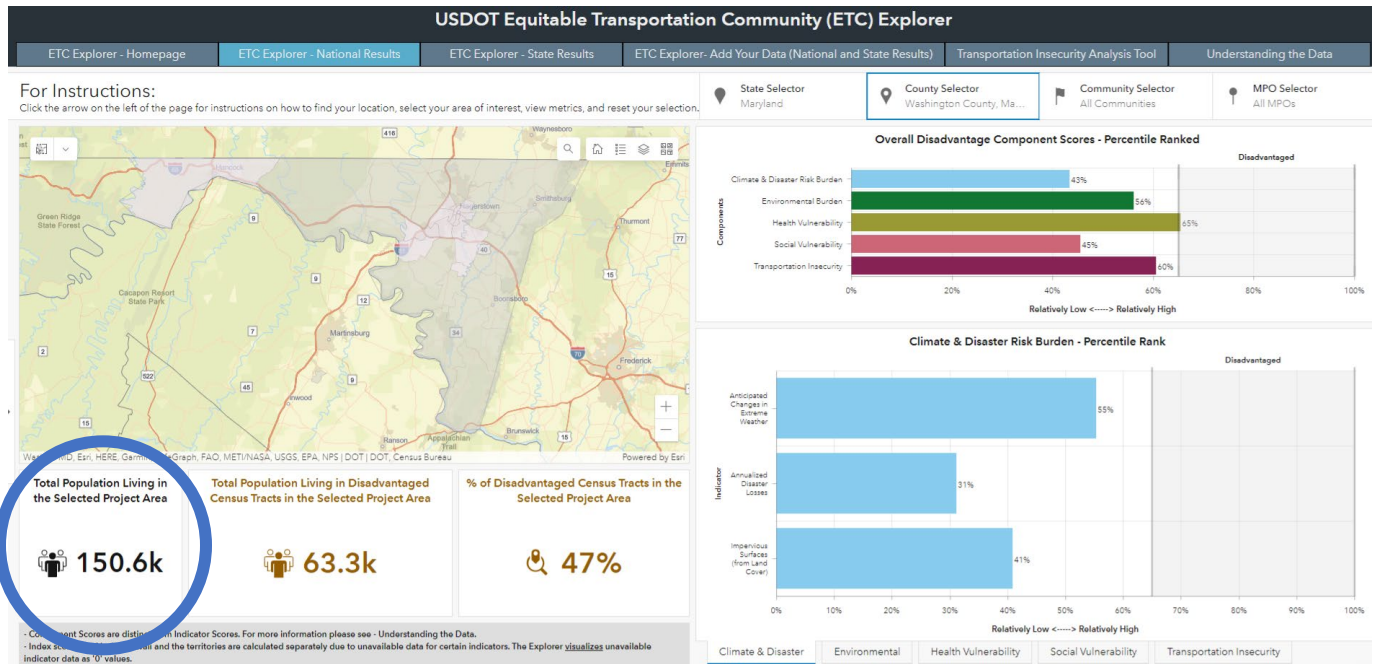
*Note that if there were no fatalities in the jurisdiction during one or more of the years, no row for that year will be shown.

Calculating Fatality Rate per 100,000 Population

Jurisdiction Population

For purposes of the FY23 SS4A application, you must use the 2020 U.S. Census American Community Survey (ACS) 5-year population estimate. For purposes of the fatality rate per 100,000 persons calculation, please include the entire Census tracts that are partially located in the jurisdiction. The SS4A program recommends using the U.S. Department of Transportation (USDOT) [Equitable Transportation Community \(ETC\) Explorer tool](#) to obtain the population to use for the fatality rate calculation. Please refer to [SS4A Calculating Percentage of Population in Underserved Communities](#) fact sheet for more information on obtaining population estimates.

- According to the ETC Explorer tool, the **jurisdiction population** for Washington County, Maryland is **150,600**.



Fatality Rate (per 100,000 population)

To calculate the fatality rate per 100,000, use the following equation:

$$\left(\frac{\text{Total Number of Vehicle Crashes}/5}{\text{Jurisdiction Population \#}} \right) * 100,000 = \left(\frac{\frac{78}{5}}{150,600} \right) * 100,000 = 10.4$$

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